Chapter 3.5

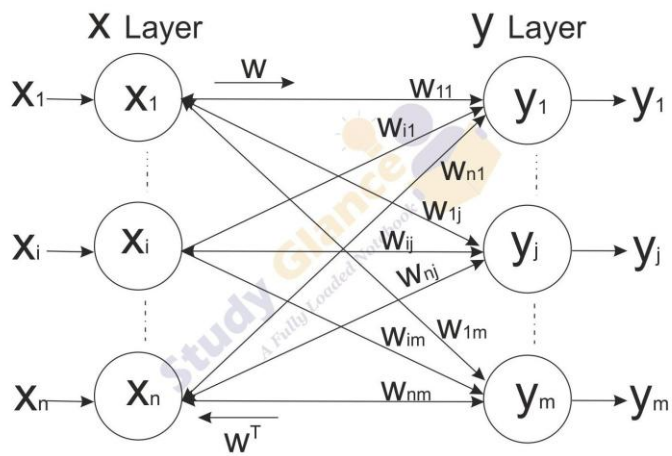
Bidirectional Associative Memory (BAM)

Topics

- Introduction
- Inputs and Outputs
- Weights and Training
- Hamming Networks
- Clustering
  - Dot product
  - Euclidean distance
  - Cost function

Introduction

- The *Bidirectional Associative Memory* (*BAM*) model has a neural network of two layers and is fully connected from each layer to the other.



- Feedback connections from the output layer to the input layer
- The weights on the connections between any two given neurons from different layers are the same
- There are no lateral connections, that is, no two neurons within the same layer are connected.
- *Recurrent* connections, which are feedback connections to a neuron from itself, may or may not be present.

Introduction (Cont.)

- The matrix of weights for the connections from the output layer to the input layer is simply the transpose of the matrix of weights for the connections between the input and output layer.

- If we denote the matrix for forward connection weights by **W**, then $\mathbf{W}^T$ is the matrix of weights for the output layer to input layer connections.

Inputs and Outputs

- The input to a BAM network is a vector of real numbers, usually in the set { –1, +1 }.
- The output is also a vector of real numbers, usually in the set { –1, +1 }
- Dimension of input and output are same or different.
- These vectors can be considered patterns
- The network makes heteroassociation of patterns.
- If the output is required to be the same as input, then you are asking the network to make autoassociation

Inputs and Outputs(Cont.)

- For inputs and outputs that are outside the set containing just –1 and +1, try following procedure.
- First make a mapping into binary numbers
- Then a mapping of each binary digit into a bipolar digit.
- Example:
- Inputs are first names of people
- Each character in the name can be replaced by its ASCII code
- Which can be changed to an binary number
- Each binary digit 0 can be replaced by –1.
- For example, the ASCII code for the letter R is 82,
- Which is 1010010, as a binary number.
- This is mapped onto the bipolar string 1 –1 1 –1 –1 1 –1.
- If a name consists of three characters, their ASCII codes in binary can be concatenated or juxtaposed and the corresponding bipolar string obtained. This bipolar string can also be looked upon as a vector of bipolar characters.

Weights and Training

- BAM does not modify weights during its operation
- The adaptive variety of BAM, called the *Adaptive Bidirectional Associative Memory*, (*ABAM)* undergoes supervised iterative training.
- A vector B is obtained at one end, as a result of C being input at the other end.
- If B in turn is input during the next cycle of operation at the end where it was obtained, and produces C at the opposite end, then you have a pair of heteroassociated vectors.
- This is what is basically happening in a BAM neural network.

Weights and Training(Cont.)

- k pairs of exemplar vectors are denoted by ($\mathbf{X_i}$, $\mathbf{Y_i}$), i ranging from 1 to k. $\mathbf{W} = X_1^T Y_1 + \ldots + X_k^T Y_k$

$$\mathbf{W}^T = Y_1^T X_1 + \ldots + Y_k^T X_k$$

- Suppose you choose two pairs of vectors as possible exemplars

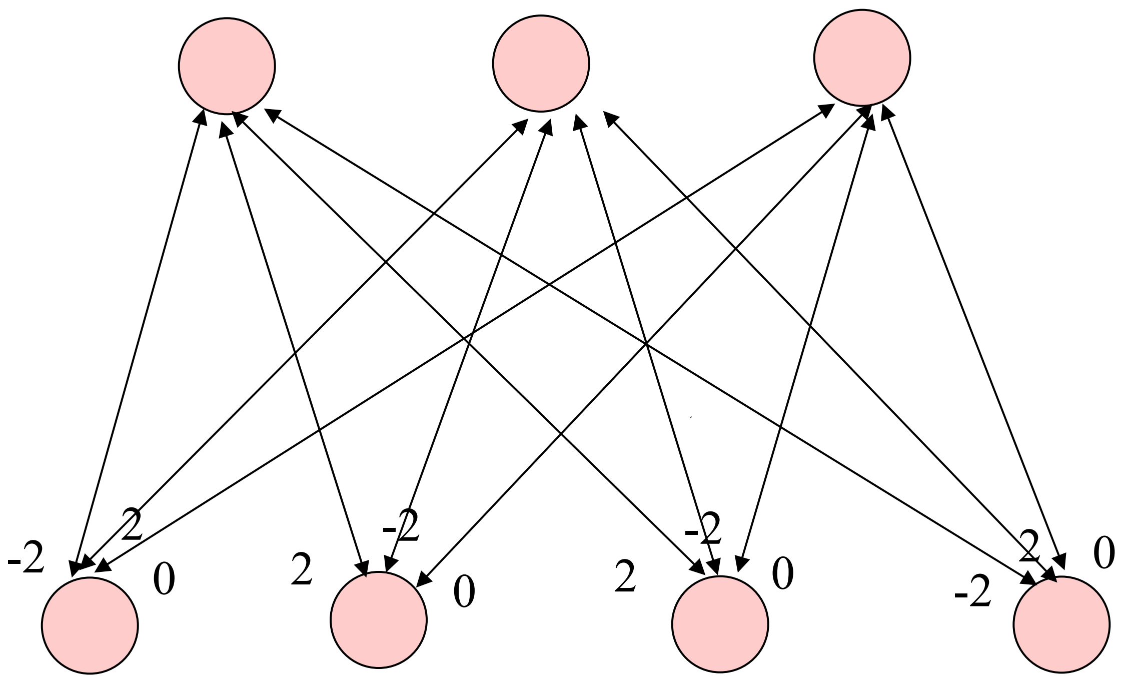
$$X_1 = (1, 0, 0, 1),\ Y_1 = (0, 1, 1)$$

$$X_2 = (0, 1, 1, 0),\ Y_2 = (1, 0, 1)$$

- These you change into bipolar components and get, respectively, (1, –1, –1, 1), (–1, 1, 1), (–1, 1,1, –1), and (1, –1, 1).

$$W = \begin{matrix} 1 \\ -1 \\ -1 \\ 1 \end{matrix} [-1\ 1\ 1] + \begin{matrix} -1 \\ 1 \\ 1 \\ -1 \end{matrix} [1\ -1\ 1] = \begin{matrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{matrix} + \begin{matrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & -1 \end{matrix} = \begin{matrix} -2 & 2 & 0 \\ 2 & -2 & 0 \\ 2 & -2 & 0 \\ -2 & 2 & 0 \end{matrix}$$

Weights and Training(Cont.)



Weights and Training(Cont.)

$$W^T = \begin{matrix} -2 & 2 & 2 & -2 \\ 2 & -2 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{matrix}$$

- The last column of *W* being all zeros presents a problem in that when the input is **X1** and whatever the output, when this output is presented back in the backward direction, it does not produce **X1**, which does not have a zero in the last component.
- There is a **thresholding** function needed here.
- The **thresholding** function is transparent when the activations are all either +1 or –1.
- When the activation is zero, you simply leave the output of that neuron as it was in the previous cycle or iteration.

Weights and Training(Cont.)

- The **thresholding** function for BAM outputs is

1 if $y_j>0$

1 if $x_i>0$

$b_j(t+1)=b_j(t)$ if $y_j=0$

$a_i(t+1)=a_i(t)$ if $x_i=0$

0 if $y_j<0$

0 if $x_i<0$

where x_i and y_j are the activations of neurons i and j in the input layer and output layer, respectively

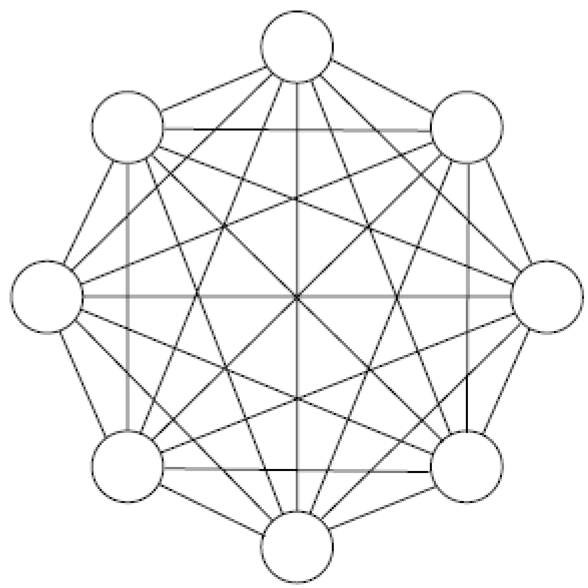
b_j (t) refers to the output of the j^{th} neuron in the output layer in the cycle t, while $a_i(t)$ refers to the output of the i^{th} neuron in the input layer in the cycle t.

Weights and Training(Cont.)

- If **X1** = (1, 0, 0, 1) is presented to the input neurons, their activations are given by the vector (–4, 4, 0).
- The output vector, after using the **threshold** function just described is (0, 1, 1). Which is **Y1.**
- The last component here is supposed to be the same as the output in the previous cycle, since the corresponding activation value is 0.
- If we feed **Y1** at the other end (B field) , the activations in the A field will be (2, –2, –2, 2),
- The output vector will be (1, 0, 0, 1), which is **X1**.
- With A field input **X2** you get B field activations (4, -4, 0), giving the output vector as (1, 0, 1), which is **Y2**.

Hopfield network

- One of the earliest recurrent neural networks reported in literature was the auto-associator independently described by Anderson (Anderson, 1977) and Kohonan (Kohonan, 1977) in 1977.
- It consists of a pool of neurons with connections between each unit i and j (i ≠ j), all connections are weighted.



Hopfield network

- The Hopfield network consists of a set of N interconnected neurons which update their activation values asynchronously and independently of other neurons.
- All neurons are both input and output neurons.
- Activation values are binary.
- The net input $s_k(t+1)$ of a neuron k at cycle t+1 is a weighted sum

$$s_k(t+1) = \sum_{j \neq k} y_j(t)\, w_{jk} + \theta_k.$$

- A simple threshold function is applied

$$y_k(t+1) = \begin{cases} +1 & \text{if } s_k(t+1) > U_k \\ -1 & \text{if } s_k(t+1) < U_k \\ y_k(t) & \text{otherwise,} \end{cases}$$

Hopfield network

- A neuron k in the Hopfield network is called stable if

$$y_k(t) = \mathrm{sgn}(s_k(t-1)).$$

- A state ∝ is called stable if, when the network is in state ∝ all neurons are stable.

Hamming Networks

- The Hamming network performs the task of pattern association, or classification, based on measuring the Hamming distance
- The network is a pattern associator of n-dimensional binary vectors labeled with m-class labels (m-class patterns).
- A new vector x' is associated with class pattern xi with the minimum Hamming distance, a Hamming distance being the number of different bits in the two patterns.
- The first layer contains connections between all the n inputs, and all the m outputs.
- The second layer is a fully connected recurrent network with m neurons
- All nodes use linear threshold activation functions.
- The first layer of connections is finding the difference between the number n and the Hamming distance between the input pattern x' and each of the m-class patterns used to calculate the connection weights in advance.

Hamming Networks(Cont.)

- The second layer of connections is calculating the maximum of these values for finding the bestmatched- class pattern x_i.

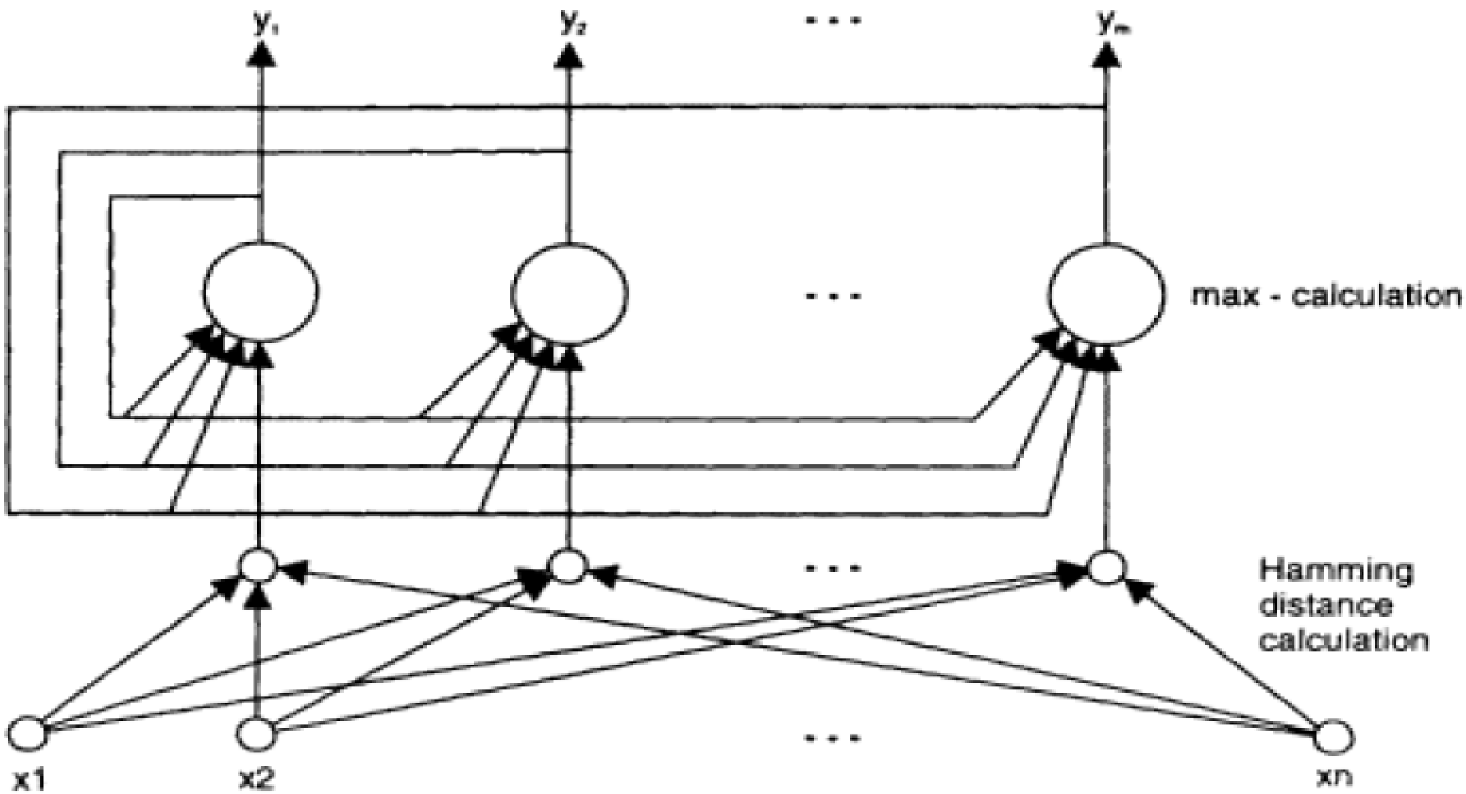


Figure 4.34
A Hamming associative neural network.

Thank you

Questions?